

MT-104

Linear Algebra

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Change of Basis

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- ▶ Given two bases $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $C = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ for a vector space V , the change of basis matrix $P_{B \rightarrow C}$ converts coordinates in basis B to coordinates in basis C .
- ▶ If $[\mathbf{v}]_B$ represents the coordinates of $\mathbf{v}$ in basis B , then:

$$[\mathbf{v}]_C = P_{B \rightarrow C} [\mathbf{v}]_B$$

Computing the Change of Basis Matrix

- ▶ To find $P_{B \rightarrow C}$, express each vector in the basis B in terms of the basis C .
- ▶ Form a matrix whose columns are the coordinates of the basis vectors of B in terms of C .

Inverse Change of Basis

- ▶ The inverse matrix $P_{C \rightarrow B}$ transforms coordinates from basis C back to basis B :

$$[\mathbf{v}]_B = P_{C \rightarrow B} [\mathbf{v}]_C$$

- ▶ $P_{C \rightarrow B}$ is the inverse of $P_{B \rightarrow C}$:

$$P_{C \rightarrow B} = (P_{B \rightarrow C})^{-1}$$

Finding a Change of Basis Matrix

Suppose we have two bases for $\mathbb{R}^3$:

$$B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \quad \text{and} \quad C = \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$$

Let:

$$B = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \right\}$$

and

$$C = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

To find the change of basis matrix, construct the augmented matrix $[B \mid C]$, where C is the 3×3 identity matrix:

$$[B \mid C] = \left(\begin{array}{ccc|ccc} 1 & 4 & 7 & 1 & 0 & 0 \\ 2 & 5 & 8 & 0 & 1 & 0 \\ 3 & 6 & 9 & 0 & 0 & 1 \end{array} \right)$$

Now perform row reduction on this augmented matrix to transform B into the identity matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 4 & 7 & 1 & 0 & 0 \\ 2 & 5 & 8 & 0 & 1 & 0 \\ 3 & 6 & 9 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\text{Row Reduction}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 1 & 0 & 2 & -3 & 1 \\ 0 & 0 & 1 & -1 & 1 & -1 \end{array} \right)$$

The matrix on the right side of the augmented matrix is the change of basis matrix P that transforms coordinates from the B basis to the C basis:

$$P = \begin{pmatrix} -1 & 2 & -1 \\ 2 & -3 & 1 \\ -1 & 1 & -1 \end{pmatrix}$$

Thus, the change of basis matrix from B to C is:

$$P = \begin{pmatrix} -1 & 2 & -1 \\ 2 & -3 & 1 \\ -1 & 1 & -1 \end{pmatrix}$$

Finding a Change of Basis Matrix Between Non-Standard Bases

Let B and C be two bases of $\mathbb{R}^3$:

$$B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$$

where

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

and

$$C = \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$$

where

$$\mathbf{c}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$$

We want to find the change of basis matrix P from basis B to basis C .

Step 1: Form the Matrix for Each Basis

Matrix B with basis vectors as columns:

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{pmatrix}$$

Matrix C with basis vectors as columns:

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

Step 2: Construct the Augmented Matrix

Combine B with C :

$$[B \mid C] = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 0 & 1 & 3 \end{array} \right)$$

Step 3: Row Reduce the Augmented Matrix

Perform row reduction to transform B into the identity matrix:

$$[B \mid C] = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 0 & 1 & 3 \end{array} \right)$$

Row Reduction

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 1 \\ 0 & 1 & 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right)$$

The matrix on the right side of the augmented matrix is the change of basis matrix P :

$$P = \left(\begin{array}{ccc} -2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array} \right)$$